Knowledge over Structure

A Controlled Study of Response Clarity Classification in Political Interviews

SemEval-2026 Task 6 (CLARITY)

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What are we classifying? Three real dev items

Clear Reply → Explicit

Sub-question: "Do you stand by your figure, 30,000?"

Answer: *"You know, I stand by the figure. A lot of innocent people have lost their life—600,000, or whatever they guessed at, is just—it's not credible. . . . "*

Why this label: he states the requested position outright.

Ambivalent → Dodging

Sub-question: "Do you see a risk of triggering a wider war?"

Answer: *"I thought you were going to ask me about the pig."*

Why this label: he ignores the question and changes the subject.

Clear Non-Reply → Declining

Sub-question: "Is it a fair assumption that the CIA operatives are in Pakistan, cooperating with the Pakistanis, and that they're sharing everything with you and vice versa?"

Answer: *"John, what's fair is, I'm not—what you must assume is that I'm not going to talk about ongoing intelligence matters. . . . "*

Why this label: he openly refuses to give the information.

- All three annotators agree on each of these. The "pig" answer is attached to all three sub-questions of its turn, so the model must work out *which part* of an answer responds to *which* sub-question.
- We predict the 9-way evasion type (Subtask 2). The 3-way clarity level (Subtask 1) follows from it.

The task: how clearly does a politician answer?

Each item pairs one **sub-question** from a journalist's turn, the **full question**, and the politician's **answer**. We predict the 9-way label and map it to the 3-way one.

Subtask 1	Subtask 2	train
Clear Reply	Explicit	30.5%
Ambivalent	Implicit, Dodging, General, Deflection, Partial/half-answer	59.2%
Clear Non-Reply	Declining to answer, Claims ignorance, Clarification	10.3%

Data and metric

- Train 3,448 rows. Dev 308 items (3 annotators). Test 237 (2 annotators, labels unreleased)
- Macro-F1 over 9 classes. A prediction counts if *any* annotator gave it
- All numbers are on dev

Why it is hard

- 69% of rows share their answer with another sub-question
- Humans disagree: one annotator vs. the other two scores **0.684**

Baselines: what the other teams did

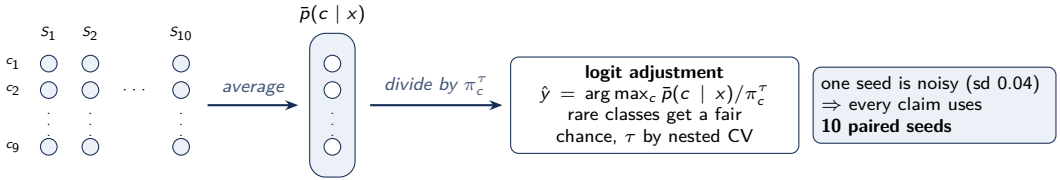
Rank S2	Rank S1	Team	Approach	S1 test macro-F1	S2 test macro-F1
1	1	**TeleAI**	DeepSeek-V3, 3 confidence-routed CoT stages, predicts S2 then maps to S1	**0.89**	**0.68**
2	7	ChulaNLP (moswisarut)	fine-tuned DeBERTa-large top-5 labels → few-shot Kimi-K2	0.82	0.61
3	9	CLaC	prompted LLMs, ahead of their fine-tuned encoders	0.80	0.59
4	4	Rasende Rakete	not described	0.83	0.59
5	6	YNU-HPCC	not described	0.83	0.59
7	3	CSE-UOI	Grok 4.1 + Gemini 3 Flash, weighted vote with difficulty gating	0.85	0.58
8	13	ttda704	structured chain-of-thought prompting	0.79	0.56
9	5	Evaluators	LoRA-tuned open LLMs	0.83	0.54
12	18	PFW (taleef)	50 DeBERTa models averaged (5 folds × 10 seeds)	0.76	0.50
17	8	tahamunawar	encoder with NLI reframing, multi-task, GNN, MIL	0.81	0.45
–	2	AsymVerify	GPT-5.2, three confidence-gated verification passes	0.85	–
–	14	SilkPeak	one DeBERTa-v3-large with joint clarity and evasion heads	0.76	–
Organiser baseline			fine-tuned Llama-70B (Thomas et al., 2024)	0.82	0.57

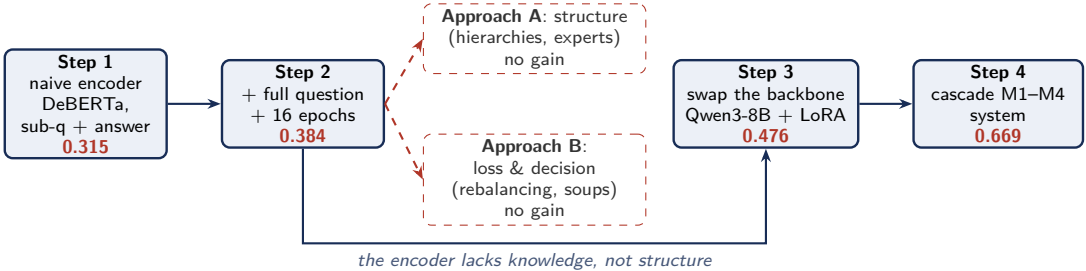
- 41 teams in S1, 33 in S2. Only 7 beat the S2 baseline, and the S2 median is about 0.52.
- The top systems are LLM pipelines that use the taxonomy. Encoders level off near **0.50 on S2**.

Test scores (237 items) from the task overview paper, Table 3. Approaches from the overview and the teams' system papers. Our numbers are on dev, so they are not directly comparable.

How the work progressed

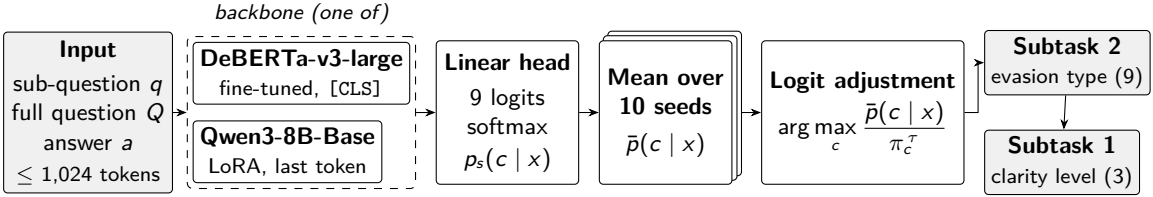
Every system: average 10 seeds, then correct for the class prior





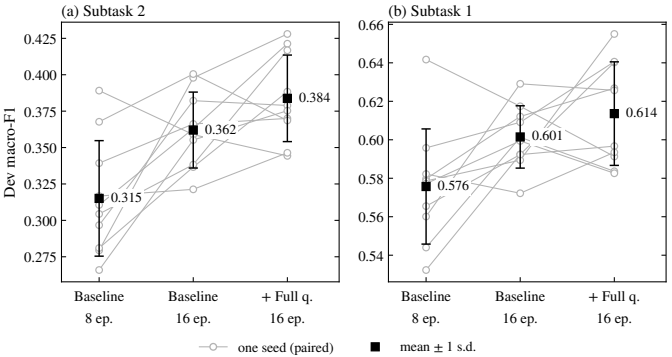
Numbers are dev Subtask 2 macro-F1 for one model (mean of 10 seeds), except Step 4, which is the full system.

The classification pipeline shared by both backbones



- **Backbone (one of two):** DeBERTa-v3-large, fully fine-tuned, reads the [CLS] state. Qwen3-8B-Base with LoRA reads the last token. Both see the sub-question q, the full question Q and the answer a ($\leq$ 1,024 tokens).
- **Head:** a new linear layer gives 9 logits and a softmax $p_s(c \mid x)$ over the evasion types for each seed s.
- **System:** average the 10 seeds into $\bar{p}(c \mid x)$, then divide by the train class prior π_c raised to τ.
- **Output:** the Subtask 2 label (9 types) fixes the Subtask 1 clarity level (3) through the taxonomy.

Phase 1: a carefully controlled encoder (DeBERTa-v3-large)

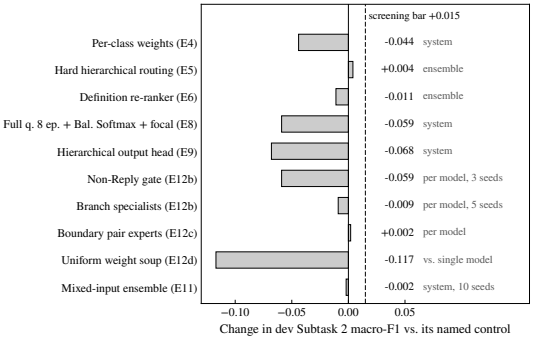


Why: our first encoder trailed ChulaNLP's (0.46), and 69% of rows share an answer, so the model needs the full question.

Result: 16 epochs + full question: **0.315 → 0.384** per model (+0.069, 9/10 seeds up). System 0.405 S2 / 0.648 S1.

Takeaway: undertraining hid the gain (at 8 epochs, 0.282 vs 0.337, loss still 1.3–1.5), so we report fit next to every score.

Encoder experiments (1/3): the decision rule and the re-ranker



E4 Per-class decision weights

Why: argmax ignores macro-F1's equal class weights and dev's different label mix.

Result: logit adjustment 0.365 → **0.438**. Nine weights reach only 0.394 (in-fold gap +0.098).

Takeaway: one scalar τ beats nine weights on 308 items. Every system uses it.

E5 Hard routing through the clarity level

Why: the organisers report that using the taxonomy hierarchy helps.

Result: 0.369 vs 0.365 for the flat model (5-seed ensemble).

Takeaway: no reliable effect. The model's own top 3 (89.6% hit) beats the taxonomy branch (81.2%).

E6 Definition re-ranker over the top 3

Why: gold is in the top 3 for 89.6% of items. A perfect chooser would score 0.745.

Result: 0.354 vs 0.365. It fixes 25 errors but breaks 37 correct answers.

Takeaway: label meaning needs knowledge that 3,400 examples cannot give. An LLM belongs here.

Encoder experiments (2/3): losses, hierarchies and ensembles

E8 Balanced Softmax + focal loss

Why: rare classes were under-predicted, and Partial/half-answer never.

Result: system 0.379 vs 0.438. Explicit F1 fell by 0.36.

Takeaway: the loss over-corrects towards rare classes. Logit adjustment does this job for free.

E9 Factorised hierarchical head

Why: post-hoc routing failed, so we trained the hierarchy into the head instead.

Result: S2 0.370 vs 0.438 (-0.068). S1 $+0.020$, within noise.

Takeaway: the flat softmax already scores all nine labels jointly.

E11 Mixed-input ensemble

Why: an exploratory mix of full-question and base models scored 0.498 on dev.

Result: on fresh seeds 5–9: 0.403 vs 0.405 for full-question models alone.

Takeaway: ideas found on dev need fresh-seed replication. We pre-register since.

All changes are measured against a named control on paired seeds. A change must first pass a **+0.015** 3-seed screen.

Encoder experiments (3/3): soups, specialists and more data

E12d Model soups

Why: one model with part of the ensemble's gain, at no extra training cost.

Result: uniform soup of 10 models 0.267 vs 0.384 per model.

Takeaway: seeds do not share a basin. Average the outputs, not the weights.

E12b/c Specialists and pair experts

Why: Non-Reply recall was 0.57, and three confused pairs held most errors.

Result: gate -0.059. Specialists $+0.018$ at 3 seeds, -0.009 at 5. Pair experts $+0.002$.

Takeaway: extra encoders re-decide what the flat model decides from the same input.

E12a Training on all of train

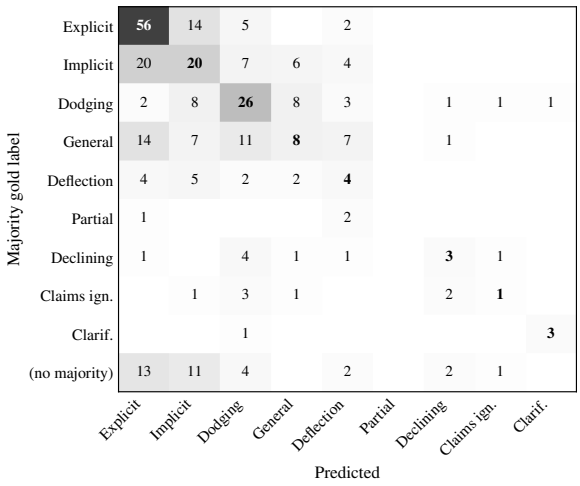
Why: picking the epoch on the slice bought little, and all data adds 11% rows.

Result: $+0.029$ S2 per model (4/5 seeds up), but S1 -0.017.

Takeaway: the only variant that helped S2, but 5 seeds cannot rank it.

Nothing trained on the same 3,448 rows moved the system. **So we changed the backbone itself.**

Where the encoder fails



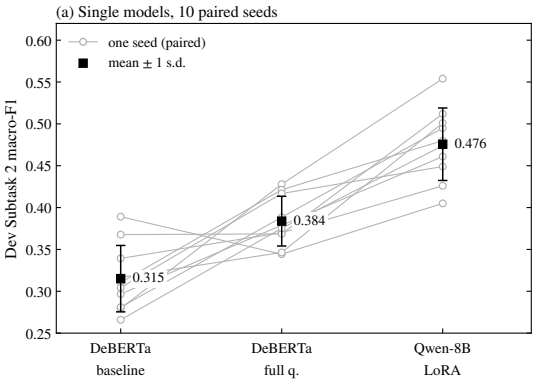
How to read it

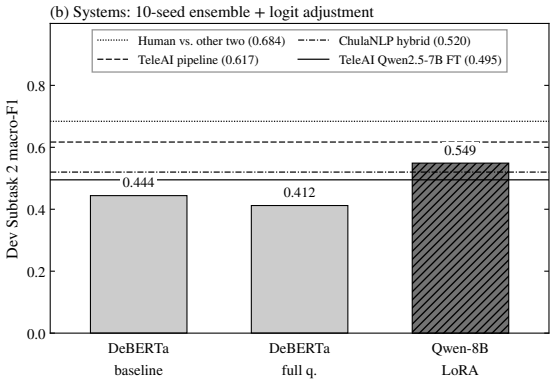
- **Majority gold label**: the label that at least 2 of the 3 annotators chose. When all three differ (33 items), the item goes to the "no majority" row.
- Columns: the final DeBERTa 10-seed ensemble's argmax. Cells are counts. The diagonal (bold) is right for 121 of 275 majority items.
- Scoring never uses the majority. A prediction counts if *any* annotator gave it.

What it tells us

- Errors sit inside *Ambivalent*: Implicit is predicted as Explicit 20 times, and General is spread over five other labels (right only 8 times).
- Partial/half-answer is never predicted.
- The encoder fails on **how committed** an answer is, the same pairs annotators disagree on.

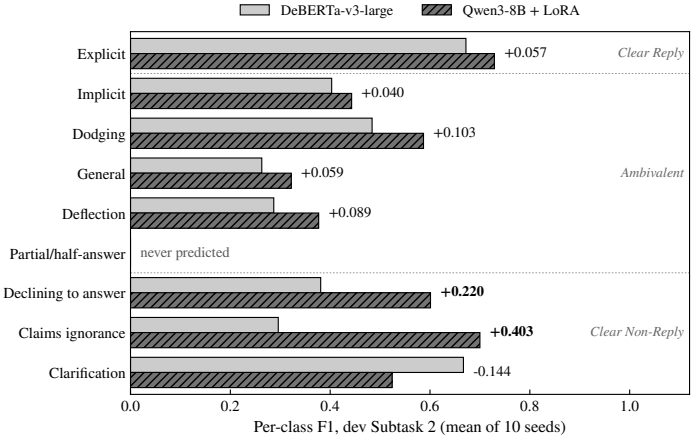
Knowledge over structure: change only the backbone





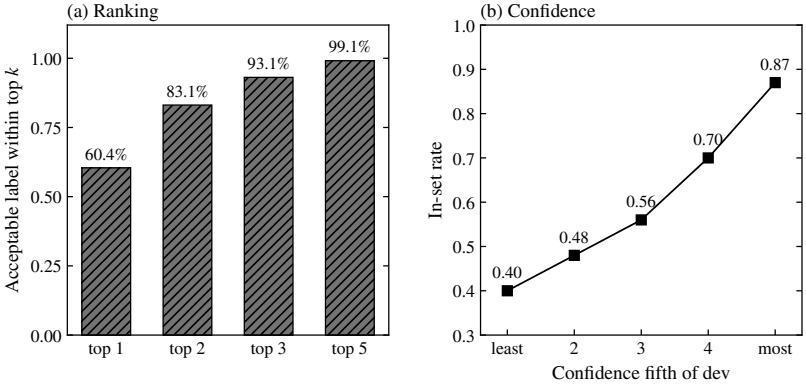
- Same rows, input and selection rule. Qwen3-8B-Base + LoRA replaces DeBERTa. Single model 0.384 → **0.476**, **10/10 seeds up**.
- Systems in (b) are means over 10 nested-CV splits. Qwen passes TeleAI's fine-tuned 7B (0.495) and ChulaNLP's hybrid (0.520).

The gain is broad, not one boundary



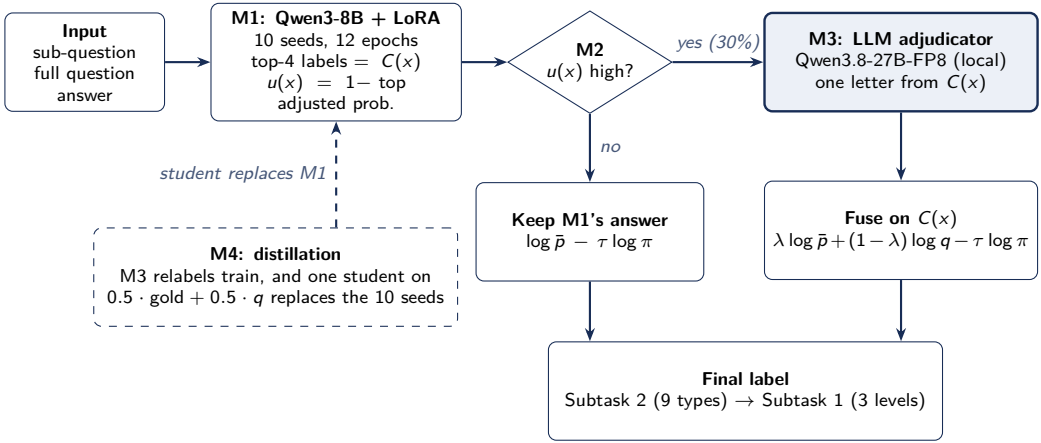
- Seven of nine classes improve, most in Claims ignorance (**+0.403**) and Declining (+0.220).
- Clarification has only 4 dev items, so its F1 swings widely (−0.144 here).
- We predicted gains on the Explicit/Implicit boundary. That prediction was refuted.

The model ranks better than it decides



- Qwen3-8B (3 epochs, seeds 0–2): an acceptable label is first for 60.4% of items but in the top 3 for **93.1%**.
- Confidence tracks correctness (in-set rate 0.40 $\rightarrow$ 0.87), so we can route only the uncertain items to a stronger chooser.

The final system: a confidence-routed cascade



- Only the least confident items reach the LLM. $\delta = 0.3$, with the threshold set on the train slice, gives **0.36 calls per item** on dev. Confident items keep M1's answer.
- Everything is chosen on train or by nested cross-validation. Dev only reports.

The four modules in brief

M1 Candidate generator

Why: Qwen ranks well but decides poorly (gold in its top 3 for 93% of items).

What: Qwen3-8B + LoRA, 12 epochs, 10 seeds, ensemble + logit adjustment. Outputs the top-4 set $C(x)$ and an uncertainty $u(x)$.

Result: **0.598** S2 / 0.743 S1 alone, with no LLM call.

M2 Uncertainty router

Why: confidence tracks correctness (in-set rate 0.40 $\rightarrow$ 0.87 across confidence fifths).

What: send only the δ least-confident items to M3. Threshold set on the train slice.

Result: $\delta = 0.3$ gives **0.36 calls/item**. Sending every item is worse (0.609 vs 0.660).

M3 LLM adjudicator

Why: a larger model knows more, but calling it on every item is costly.

What: a local 27B LLM picks one letter from $C(x)$. Its distribution q is fused with M1's.

Result: **0.669** S2 / 0.798 S1, +0.067 over M1 [+0.009, +0.124]. 33 items fixed, 9 broken.

M4 Distillation

Why: the 10 seeds are 93% of the compute at the accuracy point.

What: M3 relabels train. One student learns from $0.5 \cdot \text{gold} + 0.5 \cdot q$.

Result: student + M3: **0.608** S2 at 0.111 GPU-s per item, 4.7$\times$ cheaper than M1 alone.

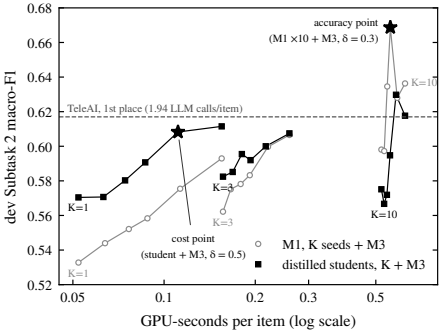
M1 + M2 + M3: the cascade on dev

δ	calls	S2	S2 (10 splits)	S1
0 (M1)	0	0.598	0.601	0.743
0.1	0.11	0.597	0.602	0.762
0.2	0.23	0.635	0.640	0.775
0.3	**0.36**	**0.669**	**0.660**	**0.798**
0.5	0.65	0.628	0.628	0.800
1	1.00	0.636	0.609	0.798

δ = target share deferred (threshold set on the train slice, realised calls in column 2).

- M1 alone (12 epochs, 10 seeds): **0.598** with no LLM call.
- At $\delta = 0.3$: **+0.067** [+0.009, +0.124]. 33 items fixed, 9 broken.
- Deferring every item is worse (0.609). The router matters.
- Screened on train: no examples, local 27B over DeepSeek V4 Flash and Gemma 4 31B. API spend $0.26.

Cost: one distilled student replaces ten seeds



Measured on one RTX PRO 6000: an M1 seed costs 0.0522 GPU-s per item, an M3 call 0.103.

M1 × 10: $10 \times 0.0522 = \mathbf{0.522}$

Accuracy point: $0.522 + 0.357 \times 0.103 = \mathbf{0.558}$ (seeds = 93%)

Cost point: $0.0522 + 0.574 \times 0.103 = \mathbf{0.111}$

$0.522/0.111 = \mathbf{4.7}\times$ cheaper than M1 × 10

The student's target (M4)

$$t(c) = 0.5 \cdot \mathbb{1}[c = \text{gold}] + 0.5 \cdot q(c)$$

- q: M3's softmax over the 4 candidate letters, 0 outside $C(x)$.
- **Why gold:** anchors the student to the annotators where M3 is wrong.
- **Why q:** says *which* other labels are plausible, knowledge a one-hot label throws away.
- The 0.5 mix was fixed in advance, not tuned.

- Student alone: 0.573 per model (+0.048, 9/10 seeds up).
- Student + M3 ($\delta = 0.5$): **0.608** S2 / 0.790 S1 at **0.111 GPU-s**, above M1 × 10 alone (0.598 at 0.522).

Final results (dev, 308 items)

System	Subtask 2	Subtask 1	LLM calls per item	GPU-s per item
DeBERTa-v3-large, 10-seed system	0.405	0.648	0	–
Qwen3-8B + LoRA, 3 epochs, 10 seeds (mid-evaluation)	0.543	0.746	0	–
M1: Qwen3-8B, 12 epochs, 10 seeds	0.598	0.743	0	0.522
Accuracy point: M1 × 10 + M3, $\delta = 0.3$	**0.669**	**0.798**	0.36	0.558
Cost point: student + M3, $\delta = 0.5$	**0.608**	**0.790**	0.57	0.111
TeleAI, 1st place (DeepSeek-V3, three stages)	0.617	0.812	1.94	–
One human annotator vs. the other two	0.684	–	–	–

- The accuracy point is above the winning system on Subtask 2 with **a fifth of its LLM calls**, and 1 output token per call.
- The cost point still beats M1 × 10 alone while costing **4.7× less** compute.

Conclusions, limitations and what comes next

What we learned

- Knowledge, not structure: changing only the backbone gained more (+0.092 per model) than all encoder-side changes combined (+0.069).
- A 27B LLM, deciding only uncertain items among four candidates, beats the winning multi-call pipeline on Subtask 2 (not yet on Subtask 1).
- Distillation turns the cascade's accuracy into a cheap single model.

Limitations

- Dev has only 308 items. The M3 gain interval is wide, and seed-subset averages suggest +0.02 to +0.04.
- Partial/half-answer (F1 0.40) and General (0.51) remain the weakest classes.

What comes next

- **Adaptive candidate sets**: let $|C(x)|$ grow on hard items instead of a fixed top 4.
- **A sharper teacher**: tune the gold/q mix and a temperature on q, and distil again.
- **Close the Subtask 1 gap** (0.798 vs 0.812) with fusion that respects the taxonomy.
- **Weak classes**: targeted examples for Partial/half-answer and General.

A small model that knows when to ask a bigger one.